

Luca Atella

Software Architect · Backend & Platform Engineering · AWS Certified

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Summary

Software architect with a backend and platform engineering background, focused on **modular systems**, **cloud-portable architectures**, **runtime governance**, and **long-lived backend platforms**.

I design systems where infrastructure abstraction, explicit runtime boundaries, operational reliability, and maintainability are part of the architecture from the beginning.

Certifications

- AWS Certified Solutions Architect - Associate
- AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner
- Cisco Networking Academy - CCNA path

Core Strengths

- Backend and platform architecture for integration-heavy systems
- AWS serverless deployment and managed-service architectures
- Cloud-portable application design and infrastructure abstraction
- Modular systems, plugin-based architectures, and runtime governance
- Docker / Docker Compose local environments and deployment automation
- IoT, edge/cloud integration, and Digital Twin architecture concepts
- Geospatial data processing and 3D terrain model generation

Professional Experience

Software Architect / Backend & Platform Engineer — Vemar S.A.S.

April 2021 - October 2024

- Designed a plugin-based backend framework for heterogeneous IoT device integrations
- Centralized business logic, authentication, API exposure, and persistence concerns
- Defined REST API contracts and backend structure
- Designed plugin loading and execution boundaries
- Introduced Docker-based development and deployment workflows
- Coordinated backend/frontend architectural decisions in a small full-stack team

Selected Projects

Cloud-Portable Fleet Management Platform

- Designed and developed a production-adopted fleet management platform with a cloud-portable architecture
- Built a FastAPI backend and contributed to React/TypeScript frontend integration
- Abstracted persistence and object storage through repository and storage contracts
- Used Docker Compose, DynamoDB Local, and MinIO for local development
- Deployed to AWS using Lambda, ECR, DynamoDB, S3, CloudFront, CloudFormation, and CloudWatch

ImportSpy (Open Source)

- Runtime contract enforcement engine for Python modules
- YAML-based declarative contracts for structural, contextual, and execution constraints
- Deterministic import-time validation model for modular and long-lived Python systems
- 15,000+ downloads on PyPI and presentation at GDG Basilicata

B3DO — Basilicata 3D Open

- Unpublished geospatial processing pipeline for transforming public Basilicata terrain datasets into 3D terrain models
- CLI-driven workflow for DTM merging, boundary extraction, raster clipping, resampling, LOD mesh generation, and textured previews
- Uses GDAL, Rasterio, NumPy, PyVista, Fiona, and Typer
- Separates raw data, processed datasets, build artifacts, and model outputs for reproducible data-to-model processing

IoT Data Aggregation Architecture for Digital Twin Systems

- Conceptual edge-to-cloud architecture for heterogeneous IoT data acquisition and aggregation
 - Includes BLE/Zigbee mesh, gateway ingestion, GraphQL API, data-specific layers, microfrontends, storage gateway, and AI-assisted processing
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Education

BSc in Computer Science and Technology (L-31) — University of Basilicata
2018 - Present

High School Diploma in Electronics, Electrical Engineering, and Automation
2013 - 2018

Languages

Italian: native · English: B1, actively improving

Tech

Python · FastAPI · React · AWS · Lambda · DynamoDB · S3 · CloudFront · CloudFormation · Docker · Docker Compose · Kubernetes · Linux · Networking · Runtime Governance · GDAL · Rasterio · PyVista